

Lateral, medial and posterior knee pain

with MARK HUTCHINSON

When I listened to other people—who seemed to have trouble even though their [knee replacements] went well—[I learned that] they didn't make the commitment to rehab . . . I can walk forever now—pain-free . . . I can hit a tennis ball again.

Billie Jean King, six years after her 2010 knee replacements

Although acute knee injuries and anterior knee pain are very common presentations in sports and exercise medicine practice, patients presenting with lateral, medial or posterior knee pain are not uncommon and can provide the clinician with both diagnosis and treatment challenges.

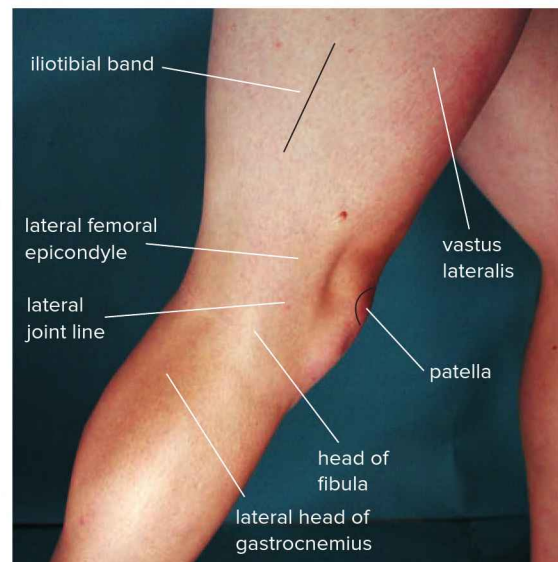
LATERAL KNEE PAIN

Athletes can present with the chief complaint of lateral knee pain for a number of underlying problems, including: lateral meniscus tears and cysts; bone bruises or cartilage injuries of the lateral compartment; patellofemoral syndrome or patellar instability; injuries of the posterolateral corner structures (popliteus, lateral collateral ligament, mid third capsular ligament anterolateral ligament); biceps insertional tendinopathy and injuries; tibiofibular joint instability; and iliotibial band syndrome at the knee.

Lateral knee pain (Fig. 37.1) is a particularly common problem among distance runners and cyclists, the most common cause of lateral knee pain being iliotibial band friction syndrome (ITBFS). Training errors as well as alignment and biomechanical abnormalities can precipitate ITBFS. Patellofemoral syndrome (Chapter 36) or instability may also present as lateral knee pain. In the older active person, lateral meniscus degeneration or lateral compartment osteoarthritis should be more strongly considered as part of the differential diagnosis. For sprinters and footballers, overuse injuries of the biceps femoris tendon and its insertion onto the fibula should be considered more highly in the differential diagnosis. The biceps femoris tendon may suffer compression and irritation as it passes posterolaterally to the knee and inserts into the head of the fibula, likely as a result of the repetitive flexion

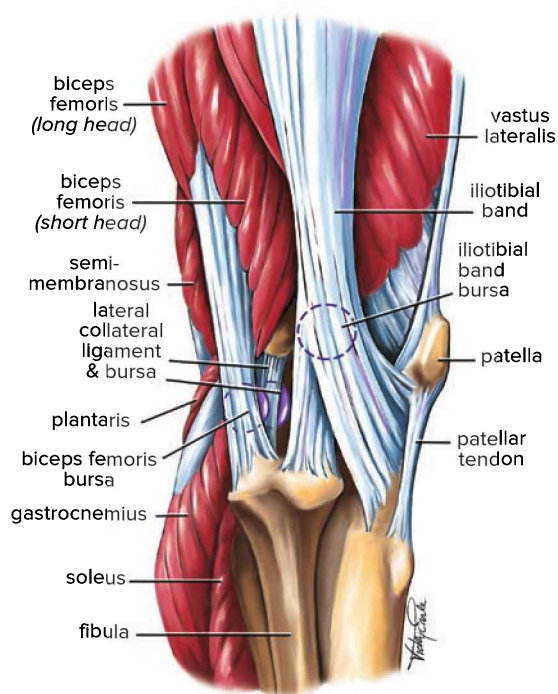
and rotation demands on the knee. Injuries of the superior tibiofibular joint may also cause lateral knee pain.

The astute clinician should always be conscious of the potential of associated injuries or referred pain involving the lateral aspect of the knee. Ligamentous injuries of the posterolateral corner are rarely isolated and frequently involve injury to the ACL or PCL. Lateral knee pain may also occur as a result of referred pain from the lumbar spine. The causes and differential diagnoses of lateral knee pain are shown in Table 37.1.



(a)

Figure 37.1 Lateral aspect of the knee (a) Surface anatomy



(b)

Figure 37.1 (cont.) (b) Anatomy of lateral aspect of the knee

Clinical approach

As with acute knee injuries (Chapter 35) and most musculoskeletal injuries, history and physical examination are key to an accurate diagnosis.

History

Most complaints involving the lateral aspect of the knee can be classified as either chronic, with no specific traumatic episode, or acute, where the athlete can recall a single event that initiated the complaint. A history of overuse or pain related to a change in training intensity suggests ITBFS or

biceps femoris tendinopathy. ITBFS has been associated with acute ramping up of training period or distance, excessive downhill running and running on an uneven surface (particularly the downside leg on a canted/sloped road). Biceps femoris strains and tendinopathy have been correlated with sprinting or kicking activities.

The pain associated with biceps femoris tendinopathy flares up on initial activity and then starts to settle after warming up, usually recurring following cessation of activity or the next day. When left untreated, pain persists during exercise and the athlete may not be able to continue with sporting activity. ITB pain usually does not settle with ongoing activity and can be associated with a snapping sensation as well as swelling and pain localised proximal to the lateral joint line.

If the athlete can remember a single episode or event, then the clinician should consider such diagnoses as a bone bruise (bone marrow lesion [BML] posterolateral corner ligament injury, proximal fibular subluxation, patellar instability or a lateral meniscus tear (Chapter 35). Lateral joint line pain associated with a sudden twisting event, accompanied by episodes of locking, is highly suggestive of a meniscus tear. The sensation of giving way may be secondary to intra-articular pathology such as a loose body, bucket-handle meniscus tear, or more commonly related to true ligamentous instability of the patella or posterolateral corner of the knee. As the population of active individuals who are over 50 years old increases, the diagnosis of lateral compartment osteoarthritis should be considered and it is possible for them to have an acute or chronic injury to a degenerative meniscus.

A complete history is necessary to avoid missing radicular or associated diagnoses. The presence of back pain may suggest referred pain from the lumbar spine. Associated neurological symptoms such as weakness and paraesthesia in the lower leg may indicate common peroneal nerve entrapment. Acute ankle injuries with

Table 37.1 Causes of lateral knee pain

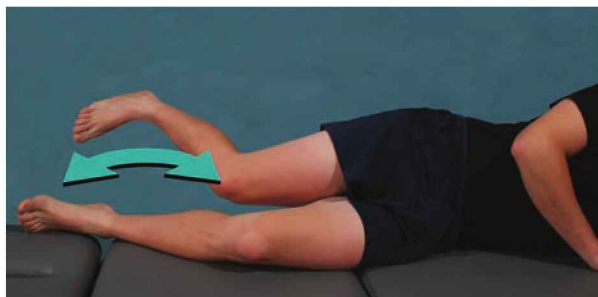
Common	Less common	Not to be missed
Iliotibial band friction syndrome Lateral meniscus abnormality • Minor tear • Degenerative change • Cyst	Patellofemoral syndrome Osteoarthritis of the lateral compartment of the knee Excessive lateral pressure syndrome Biceps femoris tendinopathy Superior tibiofibular joint sprain Posterolateral corner knee injury (Chapter 35) Synovitis of the knee joint Referred pain • Lumbar spine • Neuromechanical sensitivity	Common peroneal nerve injury Slipped capital femoral epiphysis Perthes' disease

proximal fibular pain should raise suspicion of proximal tibiofibular joint instability or a *Maisonneuve* fracture (associated fracture of the proximal fibula).

Examination

Full assessment of the ligaments of the knee (Chapter 35) should be included in the examination. Biomechanical examination should also be performed.

1. Observation
 - (a) standing
 - (b) walking
 - (c) supine
 - (d) side-lying
2. Active movements
 - (a) knee flexion
 - (b) knee extension
 - (c) repeated knee flexion (0° – 30°) (Fig. 37.2a)
 - (d) tibial rotation
3. Passive movements
 - (a) knee flexion/extension
 - (b) tibial rotation (Fig. 37.2b)
 - (c) superior tibiofibular joint
 - (i) accessory glides (Fig. 37.2c)
 - (d) muscle stretches
 - (i) ITB (Ober's test) (Fig. 37.2d)
 - (ii) quadriceps
 - (iii) hamstring



(a)

Figure 37.2 Examination of the patient with lateral knee pain (a) Active movements—repeated flexion from 0° – 30° . This may reproduce the patient's pain if ITBFS is the case. It can be performed in a side-lying position (illustrated), standing or squatting (b) Passive movements—tibial rotation. This is performed in knee flexion to assess superior tibiofibular joint movement (c) Passive movements—accessory anteroposterior glide to superior tibiofibular joint (d) Passive movement—ITB stretch. This is performed in a side-lying position with the hip in neutral rotation and knee flexion. The hip is extended and then adducted. If the ITB is tight, knee extension will occur with adduction (Ober's test)



(b)



(c)



(d)

4. Resisted movements
 - (a) knee flexion (Fig. 37.2e)
 - (b) tibial rotation
5. Functional movements
 - (a) hopping
 - (b) squat/single-leg squat
 - (c) jumping
6. Palpation
 - (a) lateral femoral epicondyle (Fig. 37.2f)
 - (b) lateral joint line
 - (c) lateral retinaculum
 - (d) lateral border of patella
 - (e) superior tibiofibular joint
 - (f) biceps femoris tendon
 - (g) gluteus medius
7. Special tests
 - (a) full knee examination (Chapter 35)
 - (i) effusion (Fig. 37.2g)
 - (ii) McMurray's test (Fig. 37.2h)
 - (b) neural tension tests
 - (i) prone knee bend
 - (ii) slump (Fig. 37.2i)
 - (c) lumbar spine (Chapter 29)
 - (d) biomechanical assessment (Chapter 8) (Fig. 37.2j)



(e)

Figure 37.2 (cont.) (e) Resisted movement—knee flexion. Concentric or eccentric contractions may reproduce the pain of biceps femoris tendinopathy (f) Palpation—lateral femoral epicondyle (g) Special test—knee effusion. Manually drain the medial subpatellar pouch by stroking the fluid in a superior direction. (1) Then 'milk' the fluid back into the knee from above (2) while observing the pouch for reaccumulating fluid (h) Special tests—McMurray's test. The knee is flexed and, at various stages of flexion, internal and external rotation of the tibia are performed (arrow). The presence of pain and a palpable 'clunk' are a positive McMurray's test and are consistent with meniscal injury. If there is no 'clunk' but the patient's pain is reproduced, then the meniscus may be damaged or there may be patellofemoral joint abnormality



(f)



(g)



(h)



(i)



(j)

Investigations

Although the majority of patients with lateral knee pain do not require investigations, MR imaging can be useful in cases of persistent lateral knee pain when a degenerative lateral meniscus is suspected. Weight-bearing radiographs with the knee slightly flexed have an improved sensitivity for osteoarthritis when compared to standard radiographs obtained in a supine position. Full-length tibia/fibula films should be obtained when associated ankle injuries are present. When imaging is not readily available, a diagnostic local anaesthetic injection can be used to differentiate local soft tissue pain (e.g. ITBFS) from intra-articular or referred pain. If the clinician is concerned for the possibility of radiculopathy from the lumbar spine, electromyography or lumbar spine imaging may be indicated.

Iliotibial band friction syndrome

Iliotibial band friction syndrome (ITBFS) is an overuse injury presenting as lateral knee pain that is exacerbated by sporting activity. It is more commonly seen in runners, cyclists, defence force recruits and endurance athletes.¹⁻⁴ Incidence rates in running range from 1.6% to 12% and 1% to 5.5% in military populations.³ In cycling ITBFS accounts for 15% to 24% of overuse injuries.³

Anatomy

Traditionally, ITBFS was considered to be the result of friction between the iliotibial band (ITB) and the underlying lateral epicondyle of the femur (Fig. 37.3) as the ITB switched from an extensor at 0–20° to a flexor at 20–90°. This friction was thought to contribute to local inflammation and irritation of an anatomical bursa lying between the tendon and the lateral femoral epicondyle (LFE).⁵ However, anatomical studies have questioned or negated this theory.^{2,4}

While both ultrasound and MRI can show thickening of the ITB over the lateral femoral condyle and often a fluid collection deep to the ITB at the same site,⁶ studies do not support the presence of an anatomical bursa around the LFE.^{2,4} Instead of a bursa, the authors claim that a richly innervated and vascularised layer of fat and connective tissue separates the ITB from the LFE and is the likely source of pain in ITBFS.^{2,4}

It is important to note that the ITB is not a discrete band but a lateral thickening of the circumferential fascia lata that envelops the whole thigh like a stocking

Figure 37.2 (cont.) (i) Special tests—slump test

(j) Special tests—biomechanical assessment Full lower limb biomechanical assessment should be performed while standing, walking and lying. Abnormal pelvic movements (e.g. excessive lateral tilt) should be noted

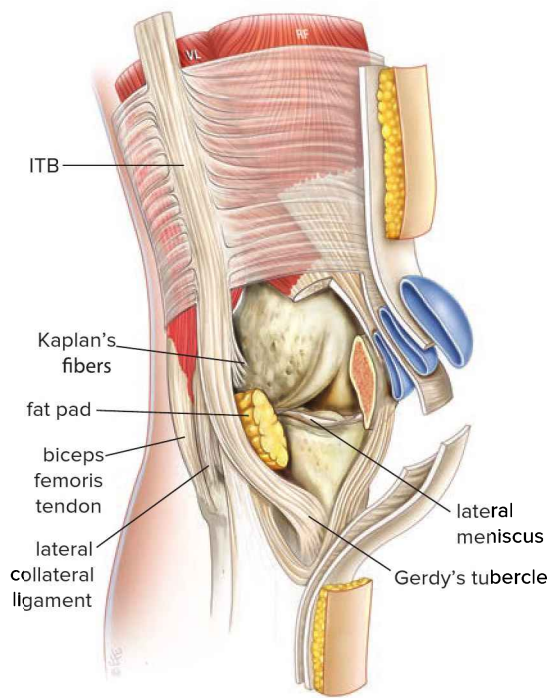


Figure 37.3 Anatomy of the iliotibial band insertion
FROM FRANKLYN-MILLER A. ET AL. *CLINICAL SPORTS ANATOMY*.
MELBOURNE: MCGRAW-HILL, 2010; P.269

(Fig. 37.3).^{1,2,4} The ITB inserts along the length of the femur down the linea aspera via the lateral intermuscular septum.^{1,2,4} Proximally, the tensor fascia lata muscle (TFL) inserts into the ITB, as does a substantial portion of the gluteus maximus muscle (GMax). The remaining portion inserts directly into the greater trochanter. Distally, the ITB crosses the LFE and is connected to the LFE by strong fibrous bands. The ITB continues, acting like a lateral ligament, from the LFE to insert onto the patella and Gerdy's tubercle on the tibia. The ITB can also project onto the fibula. This orientation demonstrates the close relationship between the ITB and the knee and hip complex and its role in lateral stability.

Clinical features

Typically, an athlete with ITBFS complains of an ache over the lateral aspect of the knee which is aggravated by running or cycling. The pain often develops at about the same distance/time during activity. Longer training sessions, downhill running or cambered courses are often aggravating factors.

On examination, tenderness is elicited over the LFE 2-3 cm above the lateral joint line (Fig. 37.2f). Crepitus and local swelling may also be felt. Repeated flexion/extension of the knee may reproduce the patient's

symptoms. Ober's test (Fig. 37.2d) often reveals ITB tightness and may produce a burning sensation. Tightness may be secondary to shortening of the TFL and/or GMax muscles proximally or excessive development of the vastus lateralis, placing increased tensile load on the ITB. Imaging is not usually required to confirm the diagnosis of ITBFS.

Biomechanics

Understanding the anatomy and pathomechanics of the ITB should assist in better targeting therapeutic interventions. Movement of the ITB around the LFE is restricted by its strong fibrous band attachments. Traditional approaches involving the alteration of tension of the fascia lata and hip musculature may not achieve the intended outcomes and can result in an increase in compressive forces around the LFE,^{2,4} leading to increased pain related to compression rather than a reduction in transverse frictional forces. The ITB clearly plays a role in stability around the hip through its tensioning effect. In hip adduction and flexion, an increase in pressure around the greater trochanter has been reported.¹ This pressure is further increased by knee flexion. The same pressure is reduced in hip abduction and knee extension. This is supported by surgical studies which have shown that ITB lengthening can lead to favourable outcomes in trochanteric bursitis presentations.¹

It was originally thought that ITBFS was associated with repetitive knee flexion and extension movement. A biomechanical study found no significant difference in sagittal knee movements between an ITBFS population and matched controls.⁵ This suggests that other planes of motion may be more relevant. For example, an increase in tibial internal rotation can augment compressive forces around the LFE by moving the ITB's distal attachment to the tibia more medially.⁷

PRACTICE PEARL

It is clear that the pathomechanics of ITBFS is complex and involves various segments along the kinetic chain. Tibial internal rotation may result from poor proximal control (increased hip adduction/internal rotation), a genu valgus strain and/or poor rearfoot mechanics.

An increase in hip adduction, especially during the loading phase of running, increases the eccentric demand on the hip abductors.⁷

Runners with ITBFS can have significant weakness of their hip abductors in the affected limb,⁸ and decreased ability of the hip abductors to eccentrically control

adduction.⁹ There is also often weakness in knee flexion and knee extension, with decreased braking forces.¹⁰ Weakness and fatigue can result in increased compressive forces around the LFE and therefore lead to ITBFS.

Neural feedback from the richly innervated fat and connective tissue between the ITB and LFE may result in decreased tension in the hip abductors to reduce these compressive loads.² This can lead to hip muscle imbalances and altered biomechanics.

Treatment

Treatment of ITBFS should address not only local symptoms, but also each segment along the kinetic chain including foot alignment and hip biomechanics to assure more favourable long-term results. A guiding philosophy is that while the pathology is felt distally, treatment should be focused proximally.

Local treatment includes cryotherapy, topical non-steroidals and modalities such as shock wave therapy¹¹ applied to the area around the LFE. Oral NSAIDs may be helpful in the initial stages to reduce pain and allow therapy.¹² Corticosteroid injection (Fig. 37.4) has traditionally been used to inject the 'bursa'.¹³ The inconsistent results from these injections can be attributed to a misunderstanding of the pathology. Whether there is a role for corticosteroid injections into the area of pain-sensitive tissue between the ITB and lateral femoral epicondyle is presently unclear. It is unlikely that they provide a cure; however, they may play a temporary role in relieving pain to allow rehabilitation. Caution should always be used when injecting steroids about ligaments for fear of weakening them.

Soft tissue treatment to the proximal ITB (Fig. 37.5a and b), dry needling (Fig. 37.5c) and self-massage with a foam roll (Fig. 37.5d) all help to reduce muscle tension

and tone on the ITB. Stretching (Fig. 37.6) is routinely proposed as a treatment technique for ITBFS. However, because the ITB inserts into the entire length of the femur via the lateral intermuscular septum, stretching exercises may have a limited effect on the ITB itself. Stretching may have some effect on reducing the tension in the tensor fascia lata (TFL).^{2, 4} Stretching of the gluteus maximus muscle may prove more beneficial because of its close relationship to the ITB.



(a)



(b)



Figure 37.4 Corticosteroid injection, if used, is aimed deep to the ITB tendon—between it and the underlying LFE

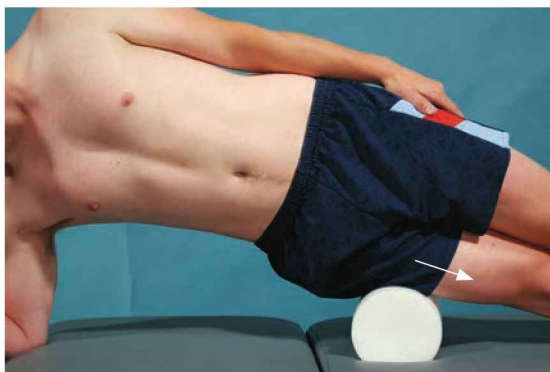
Figure 37.5 Treating tight myofascial structures
(a) Sustained myofascial tension to the proximal ITB
(b) Ischaemic pressure to the body of the ITB



(c)



(e)



(d)

Figure 37.5 (cont.) (c) Dry needling to ITB trigger points (d) Foam roll self-massage to the ITB (e) Dry needling to gluteal trigger points

Tightness of the gluteal muscles and TFL are commonly associated with ITBFS. The presence of trigger points in the TFL, gluteus minimus and gluteus medius may contribute to ITBFS. Gluteal trigger point dry needling (Fig. 37.5e) and ischaemic pressure can reduce the tension and relieve local pressure around the LFE. Strengthening of the hip external rotators (Fig. 37.7a) and abductors

(Fig. 37.7b) is an important component of the treatment of ITBFS in order to correct the underlying weakness and fatiguability of these muscles.

In rare and resistant cases, surgery may be considered. Surgery may be targeted proximally or distally. If the athlete has associated pain over both the greater trochanter and lateral aspect of the knee, then decompression/release of the ITB and excision of the 'bursa' or abnormal tissue can be done at either location or both. In patients with isolated distal pain over the lateral epicondyle, most surgeons will target the abnormal tissue by working arthroscopically beneath the ITB or via direct approach, splitting the ITB to address the abnormal tissue that lies beneath. It is essential to work aggressively with rehabilitation postoperatively as recurrence or residual pain is not uncommon.



Figure 37.6 ITB stretch—the ITB on the left side is being stretched. The symptomatic leg is extended and adducted across the uninvolved leg. The patient exhales and slowly flexes the trunk laterally to the opposite side until a stretch is felt on the side of the hip

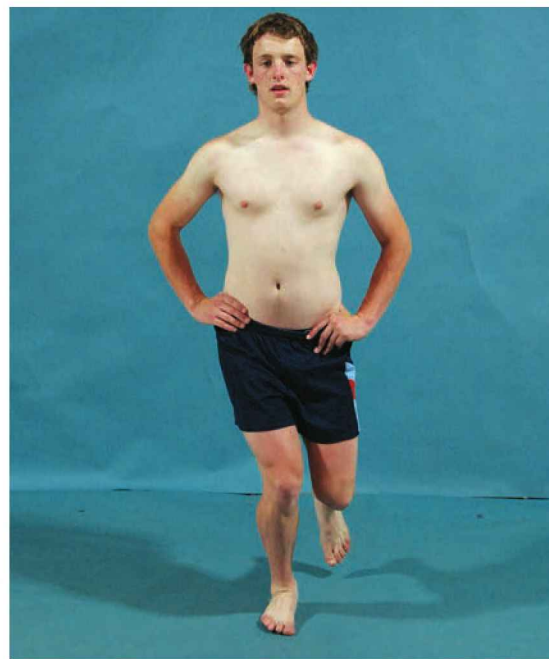
Lateral meniscus abnormality

When athletes present with lateral knee pain that is focused on the lateral joint line, lateral meniscus injury or pathology must be considered. The lateral meniscus is more circular than the C-shaped medial meniscus. It is loosely connected to the lateral joint capsule, has an unattached region through which the popliteus tendon passes and communicates with the PCL via the menisco-femoral ligaments. It is inherently more mobile than the medial meniscus. Acute meniscal injuries are discussed in Chapter 34.

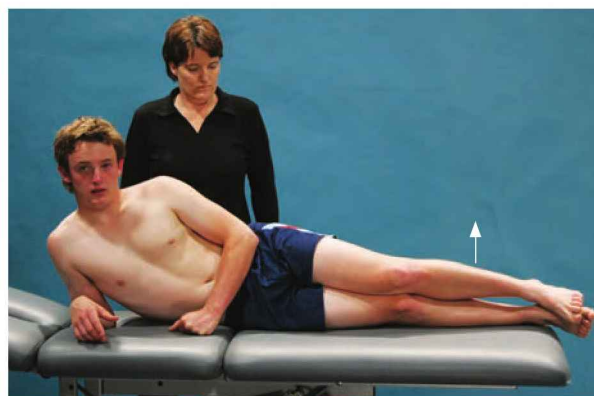
PRACTICE PEARL

In children and early adolescents, diagnosing meniscal injury can be difficult due to vague subjective findings and difficulty in localising pain.¹⁴ Meniscal injury in children is more common than previously thought and is underdiagnosed.¹⁴

A discoid (disc-shaped) lateral meniscus is an anatomical abnormality that has been reported in children. Most



(a)



(b)

Figure 37.7 Strengthening exercises for the external hip rotators and abductors (a) Exercise involves the patient standing on one leg and slowly performing a squat, maintaining pelvic stability (b) Hip abduction is side-lying

commonly asymptomatic and presenting as an incidental finding on an MRI scan,¹⁵ a discoid meniscus can also present as chronic 'snapping knee syndrome'. It is more common in the Asian population and may be bilateral. When symptomatic (i.e. painful, presence of joint effusion, symptoms of clicking and locking), it is either grossly unstable or associated with a degenerative meniscal tear, and treated arthroscopically with partial resection.¹⁵

Tenderness along the lateral joint line (2–3 cm below the lateral epicondyle and site of ITBFS) and a positive McMurray's test (pain with popping) is indicative of meniscal injury (Fig. 37.2h). Chronic degenerative meniscus tears can create a meniscus cyst which can be palpated as a soft, usually painful, bump over the lateral joint line. Radiographs can assist in ruling out other concerns, such as degenerative arthritis, osteochondritis dissecans, osteochondral loose bodies and tibial plateau fractures, within the differential diagnosis.^{14, 15} MRI is over 90% accurate for intra-articular pathology but can be associated with false positives.¹⁴

Treatment of lateral meniscus pathology is based on the athlete's age, duration of symptoms, area of meniscal pathology (vascular areas compared with non vascular) and number of associated injuries such as cruciate ligament tear. Surgeons are more likely to repair meniscus injuries that appear in young athletes or that are large and in the peripheral vascular zone. Degenerative tears in older patients are usually less repairable. The outcomes of surgical repairs have a 90% success rate when performed with a concomitant cruciate reconstruction, compared with only 60–70% success rate when the meniscus tear is an isolated injury. Symptomatic meniscus cysts are usually treated by arthroscopically debriding the intra-articular pathology.

Less common causes of lateral knee pain

Osteoarthritis of the lateral compartment of the knee

Degenerative changes in the lateral compartment of the knee may also be the source of lateral knee pain, especially in older athletes. The pain can be caused by degeneration of the articular surface of lateral tibial plateau or femoral chondyle with or without degenerative changes of the meniscus. Contributing factors include functional demand, size and depth of the lesion, varus or valgus knee malalignment and obesity.^{16, 17}

Early in the disease, the patient may give a history of increasing knee pain with activity, swelling, a grinding sensation and stiffness after a period of rest. As the disease progresses, the patient may start to experience more pain with activity, reduced function, morning stiffness or pain at night that may disturb sleep. In the early stages, examination may only reveal a small effusion or lateral joint line tenderness. McMurray's manoeuvre or varus and valgus stressing may exacerbate pain or create the sensation of grinding. A painful McMurray's is not specific for a meniscus tear unless associated popping or clicking occurs. In late stages, the grinding sensation becomes more obvious and the patient may have progressive alterations in their alignment.

The best imaging study is a weight-bearing plain radiograph in the plane of the tibial plateau with the knee slightly flexed. This has greater sensitivity than views taken with the patient supine. Other cartilage injuries that

should be considered include osteochondritis dissecans in young patients, osteochondral injuries or loose bodies following acute trauma, and spontaneous osteonecrosis of the knee (SONK) in older patients.

Initial treatment of knee osteoarthritis targets symptom relief with cryotherapy, analgesia and NSAIDs. Modification of activity and exercise prescription are helpful, particularly when combined with weight loss in overweight patients. The efficacy of oral glucosamine and chondroitin sulphate and intra-articular hyaluronic acid injections (viscosupplementation) have been debated and in some reports have a similar efficacy to NSAIDs with very little risk of serious adverse effects (Chapter 17).^{18, 19} Another alternative is the use of a deloader brace. Functional motor control assessment focusing on load distribution, such as single-leg squat, stair negotiation and gait, can identify biomechanical contributing factors.

Athletes may modify their activities to choose lower impact training (cycling and swimming rather than running) or reduced sporting challenges (doubles rather than singles tennis). Early identification of these factors may help delay further degeneration.

PRACTICE PEARL

Over the past decade, the efficacy of arthroscopic 'wash-outs' in the absence of mechanical pathology has shown no significant benefit over placebo.

Patients with severe clinical symptoms and joint space narrowing on radiographs may require unicompartmental²⁰ or, eventually, total knee replacement.

Excessive lateral pressure syndrome

Excessive lateral pressure syndrome (lateral patellar compression syndrome) occurs when there is excessive pressure on the lateral patellofemoral joint resulting from a tight lateral retinaculum. The lateral retinaculum is not one distinct anatomical structure but comprises several layers. A recent cadaveric study suggests there are three layers: a deep fascia layer (not attached to the patella), an intermediate layer (composed of the ITB and the quadriceps aponeurosis and their attachments to the patella) and the deepest layer consisting of the joint capsule.²¹ Therefore, an increase in pressure around the lateral retinaculum will affect the joint capsule, the patellofemoral joint, the ITB and the quadriceps muscles. This can lead to bone strain on the lateral patella, inflammation of the lateral retinaculum and ITBFS.

Eventually the increased strain on the lateral patella may lead to chondral degeneration or the development of a vertical stress fracture. The stress fracture can be differentiated radiologically from a congenital bipartite patella by its sharper edges with no evidence of marginal sclerosis. MRI is useful to distinguish the two as a stress injury should have associated

bone oedema.²² Chronic changes and malalignment related to chronic cartilage degeneration of the lateral patellar facet are best seen on a tangential patellar view such as a sunrise view, which allows the clinician to assess patellar tracking, patellar tilt and patellar joint space narrowing.

Initial treatment consists of patellofemoral mobilisation and soft tissue therapy to the lateral retinaculum. Taping techniques rarely help. Surgical lateral retinacular release is reserved for patients with increased, uncorrectable, lateral patellar tilt with no patellar instability. If a lateral fragment is present, removal of fragment is occasionally required.²³

Biceps femoris tendinopathy

Biceps femoris tendinopathy occurs with excessive acceleration and deceleration activities and is often associated with running and cycling. As with most tendinopathies it does not initially restrict sporting activity and therefore has a high risk of chronicity when not recognised early. Pain is described at the posterolateral corner knee, near and just proximal to the biceps insertion onto the fibular head. The pain usually settles after activity and post exercise; morning stiffness is common.

The pain is reproduced with resisted flexion, especially with eccentric contractions (Fig. 37.2e). The pain may be associated with tightness of the hamstring and gluteal muscles. Stiffness of the lumbar spine and poor core stability may also be contributing factors. The diagnosis is usually made clinically but ultrasound examination may reveal synovitis or tendinopathy.

Treatment is based on the general principles of the treatment of tendinopathy: a few days of relative rest and soft tissue therapy (Fig. 37.8) followed by progressive loading (Chapters 4 and 17). Strengthening of the hamstrings (Fig. 37.9) and the hip muscles, specifically the gluteus maximus, may also prove useful.²⁴ (See also Chapter 31 for the management of lateral hip pain.)

If prolonged pain and reduced function affect sporting activity despite quality conservative measures, surgical approaches may be considered. These include stripping of the paratenon, removal of degenerative tissue and other repair techniques for torn tendons.²⁵

Superior tibiofibular joint injury

Injury to the superior tibiofibular joint (STFJ) is a rare but commonly missed cause of lateral knee pain. The STFJ is comprised of the articulation between the lateral condyle of the tibia and the fibular head. The joint capsule is strengthened by superior, anterior and posterior ligaments.^{26, 27} The STFJ plays a role in tibiofemoral joint stability through its surrounding anatomical structures, which include the lateral collateral ligament, arcuate ligament, popliteofibular ligament, biceps femoris tendon and popliteus muscle.²⁷ The STFJ externally rotates



Figure 37.8 Soft tissue therapy in the treatment of biceps femoris tendinopathy. Ischaemic pressure at the musculotendinous junction (shown) and muscle belly can be effective



Figure 37.9 Eccentric strengthening exercises in the treatment of biceps femoris tendinopathy. Drop-and-catch is performed in prone positions. This may be progressed to include hip flexion (i.e. patient lying over the end of the bed)

during ankle dorsiflexion and is thought to dissipate torsional stresses from the ankle.²⁷ When injured, the STFJ can affect knee and ankle function and can be a source of lateral knee pain.

Clinical assessment

STFJ injury may result from twisting injuries or direct trauma resulting in capsuloligamentous sprains, subluxation or dislocation of the fibula. The mechanism of injury is often described as a combination of rotation and knee flexion (e.g. pivoting, cutting) and has been reported in sports such as rugby, soccer, skiing and the various forms of martial arts.²⁵ Additionally, STFJ injuries can be secondary to severe ankle injuries (a Maisonneuve variant) (Chapter 41). In the older athlete, osteoarthritis of the STFJ can accompany knee arthritis and can be confirmed by careful review of radiographic imaging.²⁶

In the presence of an acute dislocation, the athlete may present with a prominent head of fibula, lateral knee pain and swelling around the STFJ. The athlete may report popping or clicking. Pain may be exacerbated by ankle movements and weight-bearing activities. It is important to immediately assess and rule out a peroneal nerve injury. STFJ injuries can be associated with popliteus tendon injury and injuries to the posterolateral corner of the knee.²⁶ Early recognition of posterolateral corner injuries is essential to optimising clinical and surgical outcomes for the patient (Chapter 35).²⁶

In the presence of a sprain, pain may be local or may refer distally along the lateral calf. The STFJ is often tender, especially directly over the head of the fibula, and passive movement of the joint may be restricted by hamstring (biceps femoris) and peroneal muscle spasm. With these muscles relaxed, excessive movement may be detected on passive gliding (Fig. 37.2c).

Treatment

Acute and painful subluxations of the tibiofibular joint can be managed conservatively with immobilisation (approximately 2–3 weeks), followed by a progressive strengthening program of the hamstring and calf muscles.²⁵ Excessively mobile joints are more difficult to treat and can be associated with generalised hypermobility, especially in the female athlete. Predisposing factors, such as excessive pronation, which place greater torsional forces through the joint, should be assessed and may require correction. Dislocated joints require reduction with possible stabilising fixation.

Manual mobilisation can be an effective treatment for the stiff tibiofibular joint. Local electrotherapy may help relieve pain, but care must be taken as the peroneal nerve is quite close. Strengthening of the muscles around the knee, especially the tibial rotators (Fig. 37.15), may help stabilise the joint, reduce pain and improve function.

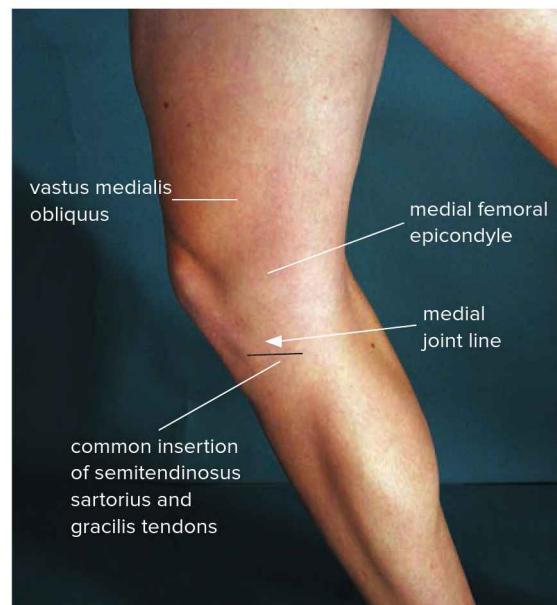
Occasionally a corticosteroid injection may be used. For patients with chronic pain or instability which impairs function, surgical options include arthrodesis and fibular head resection.²⁸ Proximal tibiofibular joint capsule reconstruction using a gracilis autograft can lead to subjective improvements and could aid return to sport.²⁵ However, strength may continue to be reduced.

Referred pain

Pain may refer from the lumbar spine to the lateral aspect of the knee. Referred pain is usually a dull ache and is poorly localised. The slump test may be positive. The lumbar spine should be examined in athletes presenting with atypical lateral knee pain.

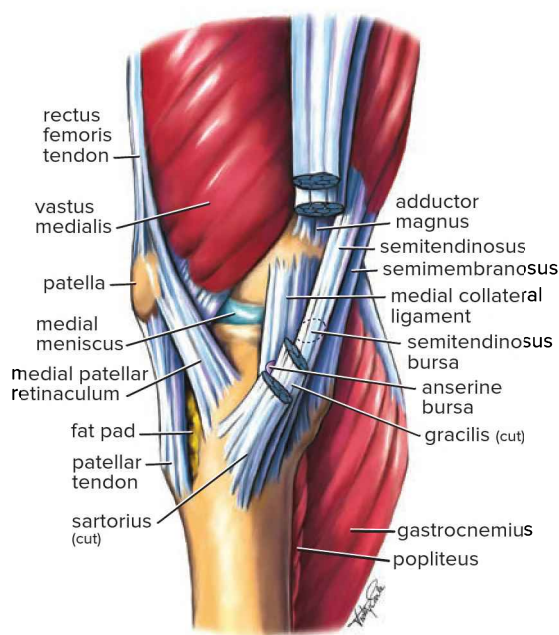
MEDIAL KNEE PAIN

Like the lateral aspect of the knee, being able to evaluate and treat medial knee pain begins with an understanding of the basic anatomy and function of the structures on the medial aspect of the knee (Fig. 37.10). The medial collateral ligament (MCL) provides stability versus valgus stress of the knee and originates at the medial epicondyle and travels beneath the pes tendons to insert 10–15 cm distal to the medial joint line on the tibia. The sartorius, semitendinosus and gracilis tendons (pes anserine) serve to flex the knee and insert just anterior and proximal to the tibial insertion of the MCL. The medial patellofemoral ligament (MPFL)



(a)

Figure 37.10 Medial aspect of the knee (a) Surface anatomy



(b)

Figure 37.10 (cont.) (b) Anatomy

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is a key stabilising structure for patellar stability and runs from the supero-medial border of the patella via the medial retinaculum to a point posterior to the femoral insertion of the MCL (Fig. 35.17). The medial meniscus is intimately attached to the MCL and may be injured with MCL injuries. Ultimately, injury to any of these structures can cause acute or chronic medial knee complaints in athletes. The causes of medial knee pain are shown in Table 37.2.

Patellofemoral syndrome

Anteromedial knee pain is common and is usually associated with patellofemoral syndrome or patellar mal-tracking. Patellofemoral syndrome is discussed in Chapter 36; however, some discussion is included here as it relates to the differential diagnosis of medial knee pain. Indeed, recent studies suggest that the MPFL may play a significant role in medial knee pain associated with patellofemoral syndrome.^{29, 30}

When an athlete dislocates his or her patella, he or she will invariably injure the medial retinaculum and MPFL. Most of these will be injured in their mid-substance or off the femur; therefore, in addition to the sensation of apprehension and palpable effusion, the most common finding on examination is pain to palpation over the MPFL and medial retinaculum.

In chronic cases, pain over the MPFL and medial retinaculum may persist due to recurrent instability. Chronic inflammation of the medial retinaculum due to recurrent instability or repetitive irritation from knee flexion may present as a symptomatic plica (Chapter 36). Non-surgical treatments may include cryotherapy, taping, patellofemoral bracing and steroid injections. Surgical options can include arthroscopic resection of a resistant medial plica or medial patellofemoral ligament reconstruction targeting both pain and recurrent instability.³⁰

Medial meniscus abnormality

Pain that is localised over the medial joint line is usually secondary to either meniscus tear or degeneration, or a cartilage injury in the medial compartment of the knee. In the young adult patient, a small tear of the medial meniscus may cause a synovial reaction and medial knee pain (Fig. 37.11). In the older patient, gradual degeneration of the medial meniscus can present as gradual-onset medial knee pain. The clinical scenario is similar to that described earlier for a lateral meniscal abnormality.

Table 37.2 Causes of medial knee pain

Common	Less common	Not to be missed
Patellofemoral syndrome Medial meniscus • Minor tear • Degenerative change • Cyst Osteoarthritis of the medial compartment of the knee	Synovial plica (Chapter 36) Pes anserinus • Tendinopathy • Bursitis Medial collateral ligament • Grade 1 sprain/bursitis • Pellegrini-Stieda lesion Referred pain • Lumbar spine • Hip joint • Neural mechanosensitivity	Tumour (in the young) Slipped capital femoral epiphysis Referred pain from the hip Perthes' disease

The athlete is generally aged over 35 years and complains of clicking and pain with certain twisting activities, such as getting out of a car, rolling over in bed, changing direction, and cutting and pivoting movements during sporting activity. Examination reveals joint line tenderness and a positive McMurray's test, that is, pain with a pop (Fig. 37.2h). MRI is the investigation of choice.

Like the lateral meniscus, treatment for medial meniscal injuries depends on the patient's age, chronicity, tear pattern, functional demands and associated ligamentous injuries (see Chapter 35). Conservative management is warranted for non-displaced or degenerative tears; however, if this fails, surgical intervention may be required. Complete meniscectomies are contraindicated as the remaining uninjured portion of the medial meniscus can continue to contribute to shock absorption.³¹ Complete

meniscectomies are associated with osteoarthritis. Assuming an appropriate tear pattern, meniscus repair is preferred, especially in younger athletes.

Osteoarthritis and chondral injuries of the medial compartment of the knee

Osteoarthritis (OA) of the knee generally affects older athletes (over the age of 50 years), but has been reported in adolescents and young adults.^{32, 33} It is important to remember that OA affects not only articular cartilage, but also subchondral bone and synovium. Ligamentous, meniscal and chondral injury, as well as repetitive joint loading, can predispose the knee joint to OA.^{32, 33}

Epidemiology

Risk factors for developing premature arthritis include obesity, sports participation, previous knee injury and genetic predisposition.^{32, 33} Increasing obesity rates in children, specifically a body mass index (BMI) above the 95th percentile for height and weight ratios (for 2-19 year olds), has been linked to increased risk of musculoskeletal injury.^{32, 33} Athletes participating in power sports, such as wrestling, boxing, weightlifting, and team sports, such as soccer, basketball and football, have a higher incidence of premature knee arthritis.^{32, 33} However, some sports have a lower incidence of OA. These sports include cross-country skiing, walking and swimming.³²

PRACTICE PEARL

In the early stages, it is very difficult to distinguish medial compartment OA clinically from medial meniscal injury.

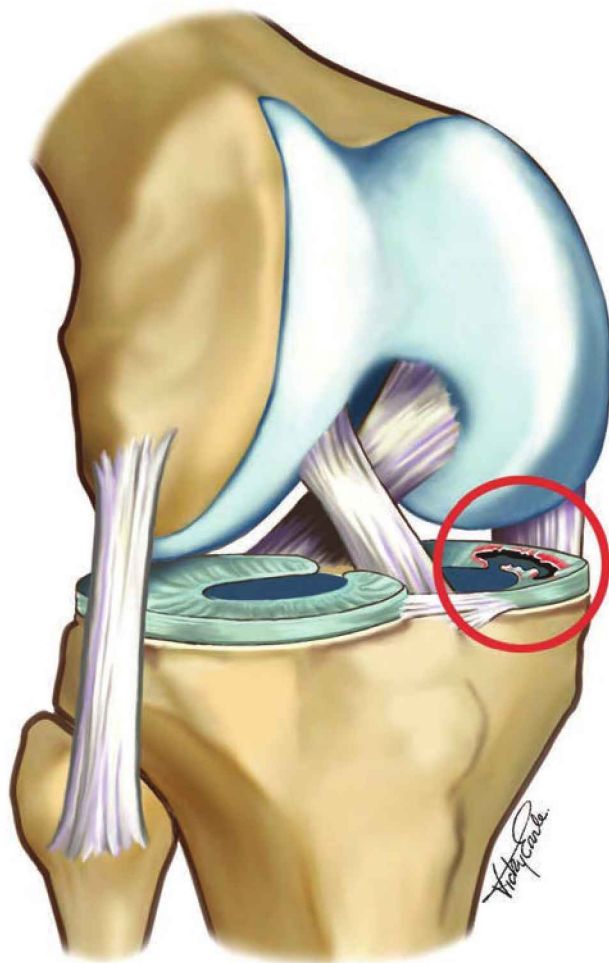


Figure 37.11 Medial meniscus tear—a common cause of medial knee pain

A painful McMurray's manoeuvre may be present in either case. The sensation of grinding is consistent with arthritis while a palpable clunk during the McMurray's manoeuvre is more indicative of a meniscus tear. If weight-bearing radiographs are normal, MRI can distinguish the two conditions using specific articular cartilage sequences, including the technique of delayed gadolinium-enhanced MRI of cartilage (dGEMRIC), which estimates cartilage quality.³⁴

Identifying and monitoring athletes at a high risk of OA can allow the opportunity for prevention and disease modifying interventions. Assessment of family history, previous knee injuries, muscle imbalances (specifically quadriceps weakness), lower limb biomechanics and body weight can be incorporated into a screening assessment. Preventive lower-limb-specific rehabilitation programs reduce the risk of lower limb injuries.³⁵⁻³⁷

Treatment

Initial management of osteoarthritis includes symptomatic relief with cryotherapy, analgesia and NSAIDs.¹⁹ A randomised clinical trial found that a custom-made, valgus-producing functional knee (unloader) brace provided significant benefit in this population of patients aged in their 60s.³⁸ Biomechanical studies have shown that these braces maintain condylar separation.³⁹ The benefits of glucosamine and chondroitin sulfate supplementation and viscosupplementation injections have been debated but do not appear to have a deleterious effect other than cost (Chapter 17).^{18, 40} Modification of activity and exercise prescription, together with weight loss, should be addressed early.

If clinical symptoms are persistent and severe, referral to an orthopaedic surgeon is warranted. Arthroscopic 'clean out' procedures for patients with arthritis but no mechanical pathology have fallen into disfavour but may be effective if mechanical symptoms are present. Surgical intervention may include high tibial osteotomy, unicompartmental replacement and total knee replacement. Interestingly, early surgical intervention to prevent OA has not been shown to be superior to non-surgical treatments;³² indeed, some surgeries have been linked to the development of early degenerative changes in young adults, for example, ACL reconstruction.^{41, 42}

Less common causes of medial knee pain

Pes anserinus tendinopathy/bursitis

Athletes who present with pain over the anteromedial aspect of the knee distal to the joint line should be evaluated for a stress fracture or pes anserine bursitis. A stress fracture would present with focal pain and pain with impact. Pes anserine bursitis is characterised by localised tenderness and swelling three fingerbreadths distal to the medial joint line but it can mimic medial meniscus pathology. Active contraction or stretching of the medial hamstring muscles can reproduce pain.

The pes anserinus ('goose's foot') is the combined tendinous insertion of the sartorius, gracilis and semitendinosus tendons at their attachment to the tibia (Fig. 37.10b). The area is richly innervated as each muscle (sartorius, gracilis, semitendinosus) is supplied by a different nerve (femoral, obturator and tibial respectively). The primary action of these muscles is to flex the knee; however, they play an important stability role by resisting valgus strain.^{43, 44} The risk of injury to the pes anserinus is increased with excessive valgus stress. Excessive valgus stress can be the result of collateral ligament instability, meniscal pathology, muscle imbalances or valgus knee deformity.⁴⁴ The pes anserinus bursa lies between the pes anserinus insertion and the periosteum and may become inflamed as a result of overuse. The pes anserinus tendons themselves can also be compressed or irritated, resulting

in a tendinopathy. Pes anserinus tendinopathy/bursitis is uncommon but may occur in swimmers (particularly breaststrokers), cyclists or runners.

Initial treatment follows the general principles of tendinopathy/bursitis management including cryotherapy, NSAIDs, electrotherapeutic modalities and eccentric stretching/strengthening. Ultrasound-guided corticosteroid injection into the bursa can be effective in reducing pain.⁴⁵ Advanced imaging such as MRI is rarely necessary, but may clarify the diagnosis or identify concomitant pathology in resistant cases.⁴⁵ For long-term results, biomechanical factors including the core, foot alignment and muscle imbalances, which can influence valgus forces, need to be addressed.

Chronic MCL injuries and Pellegrini–Stieda syndrome

Due to its excellent vascularity and multiple-layered anatomy, the medial aspect of the knee and in particular the MCL has an excellent healing potential with conservative treatment. Nonetheless, a few factors correlate with chronic medial knee pain due to resistant healing of the MCL.

Chronic repetitive valgus loading of the knee has been correlated with medial knee pain in breaststrokers. The repetitive whip kick can stress the MCL or nearby bursa. Treatment is usually symptomatic with rest, kick modifications, cryotherapy, NSAIDs followed by a gradual return when asymptomatic. Steroid injections are discouraged as they may weaken the MCL.

Pellegrini–Stieda syndrome is a disruption of the femoral origin of the MCL with calcification at the site of injury. It can be a difficult lesion to assess before radiological changes become evident. It may follow direct trauma or, less frequently, a grade 2 or 3 sprain of the MCL. Note that imaging abnormalities exist in asymptomatic individuals.⁴⁶

PRACTICE PEARL

Pellegrini–Stieda syndrome is an important cause of knee stiffness. The patient complains of difficulty straightening the leg and twisting.

Examination reveals marked restriction in joint range of motion with a tender lump in the proximal portion of the MCL. Treatment consists of active mobilisation of the knee joint and infiltration of a corticosteroid agent to the tender MCL attachment if pain persists. Surgery may be indicated in the presence of significant bone formation and persistent symptoms (pain and stiffness).⁴⁷

Injuries of the distal insertion of the MCL can be difficult to distinguish between the pes anserine insertion and a chronic MCL injury. In rare cases, the distal MCL is avulsed as a sleeve off its tibial insertion, leading to persistent pain and instability. Clinically, the patient

will have pain 3–4 fingerbreadths distal to the joint line on the medial aspect of the knee. The key to clarifying the diagnosis with pes anserine bursitis is that a chronic injury of the MCL will be accompanied by valgus knee instability. If symptoms have been present for longer than 2 months, the potential for healing is poor and surgical repair is usually required.

Other causes of medial knee pain

Perhaps the most important thing to consider in an athlete with medial knee pain is not the knee at all. The most important thing is to consider the pathologies that can radiate to the knee and, if missed, lead to catastrophic outcomes.

For the skeletally immature athlete with medial knee pain, the clinician should always consider the possibility of a hip pathology such as slipped capital femoral epiphysis or Legg-Calvé-Perthes' disease (Chapter 44). For skeletally mature athletes, one must be aware that a femoral neck stress fracture (Chapter 32) can radiate pain to the medial knee. Since missing these diagnoses has potential catastrophic long-term implication on hip function, every athlete with medial knee pain should be screened with a simple hip examination assessing range of motion, looking for these associated pathologies. Radicular pain to the medial knee may also be caused by lumbar spine pathology such as a herniated disc but any potential catastrophic implication is usually more obvious, with associated paraesthesias or motor weakness.

POSTERIOR KNEE PAIN

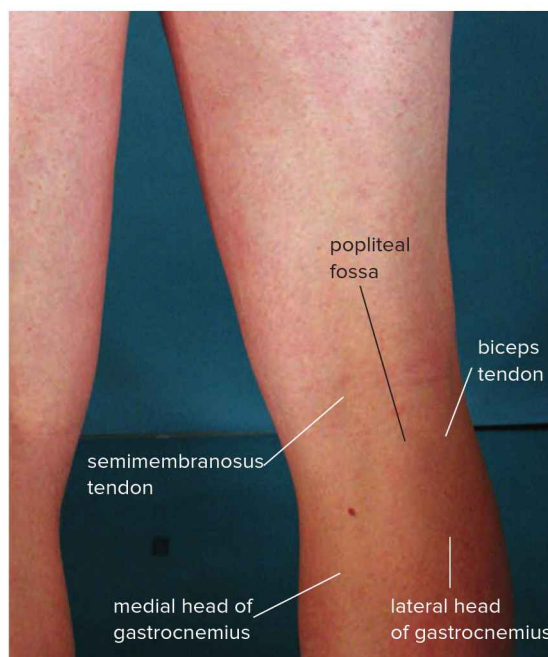
The posterior knee is the most difficult and challenging region of the knee in which to make an accurate and specific diagnosis (Fig. 37.12). The anatomy is complex, there are multiple possible pathologies, structures are less superficial and the posterior knee pain is a common site of referred pain from the lumbar spine and from the patellofemoral joint. Reactive changes from other pathologies such as a knee effusion related to a meniscus tear may lead to pain and tightness of the back of the knee. The causes of posterior knee pain are shown in Table 37.3.

Clinical evaluation

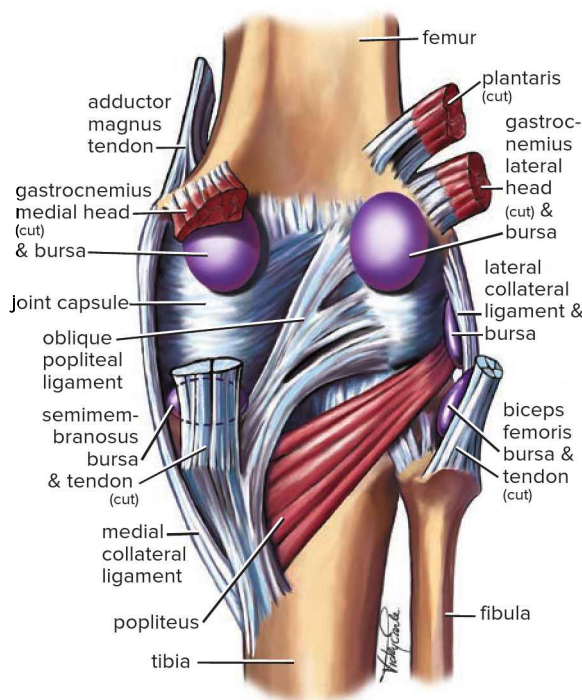
Posterior knee pain precipitated by acceleration or deceleration (e.g. downhill running, kicking, sprinting) may trigger symptoms related to biceps femoris or popliteus tendinopathy.

History

Pain described as a poorly localised, dull ache that does not directly relate to activity is more suggestive of referred pain. The presence of low back pain, patellofemoral symptoms or other knee pathology may provide diagnostic clues.



(a)



(b)

Figure 37.12 Posterior aspect of the knee (a) Surface anatomy (b) Anatomy

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Table 37.3 Causes of posterior knee pain

Common	Less common	Not to be missed
Knee joint effusion Baker's cyst Referred pain • Lumbar spine • Patellofemoral joint • Neural mechanosensitivity Biceps femoris tendinopathy	Popliteus tendinopathy Gastrocnemius tendinopathy	Deep venous thrombosis Claudication Posterior cruciate ligament sprain

Examination

In the examination of the posterior aspect of the knee, it is important to differentiate between local and referred causes of pain. The slump test may indicate whether the pain is referred from the lumbar spine or neural structures. It is also important to detect the presence of an effusion as this may be the cause of the posterior knee pain.

1. Observation
 - (a) standing (Fig. 37.13a)
 - (b) prone
2. Active movements
 - (a) flexion
 - (b) extension
 - (c) tibial rotation
3. Passive movements
 - (a) flexion
 - (b) extension (with adduction and abduction)
 - (c) tibial rotation
 - (d) muscle stretch—hamstrings
4. Resisted movements
 - (a) knee flexion
 - (b) knee flexion in external tibial rotation (Fig. 37.13b)
 - (c) external tibial rotation (Fig. 37.13c)
5. Palpation (Fig. 37.13d)
 - (a) hamstring tendons
 - (b) popliteus
 - (c) joint line
 - (d) gastrocnemius origin
6. Special tests
 - (a) knee effusion
 - (b) examination of knee joint (Chapter 35)
 - (c) neural tension—slump test (Fig. 37.2i)
 - (d) examination of lumbar spine (Chapter 29)
 - (e) biomechanical examination (Chapter 8)
 - (f) squat

Figure 37.13 Examination of patient with posterior knee pain (a) Observation—standing. Obvious swelling or fullness of the posterior aspect of the knee joint suggests a Baker's cyst. Inspection may reveal a biomechanical abnormality (b) Resisted movements—knee flexion in external tibial rotation. Resisted contraction of the popliteus tendon



(a)



(b)



(c)



(d)

Figure 37.13 (cont.) (c) Resisted movement—popliteus. With the patient supine, hips and knees flexed to 90° and the leg internally rotated, the patient is asked to 'hold it there' while the examiner applies an external rotation force (d) Palpation. This should be performed with the knee in flexion. Tenderness can be elicited over the hamstring tendons (shown), gastrocnemius origin or popliteus. It is helpful for the patient to gently contract and relax individual muscles that are being palpated in order for the examiner to precisely pinpoint the site of pain

Investigations

Ideally, a good history and physical examination should lead to a confident diagnosis and obviate the need for additional investigations or advanced imaging. Nonetheless, it must be acknowledged that the anatomy and potential pathologies in the posterior aspect of the knee are

complex and additional studies may be prudent to clarify the diagnosis. Since many patients with deep venous thrombosis are asymptomatic, the clinician should be readily willing to order venous Doppler studies if history and symptoms dictate. Radiographs may be reserved for resistant cases, but may reveal loose bodies in the posterior compartment of the knee. Diagnostic ultrasound may be used to confirm the presence of a Baker's cyst or identify a tendinopathy, and is a valid tool to guide targeted injections. In most cases, MRI is the investigation of choice if the initial diagnosis does not respond to treatment.

Baker's cyst

A Baker's cyst (popliteal cyst) is a synovial fluid filled mass located in the popliteal fossa.⁴⁸ The mass is often an enlarged bursa located beneath the medial gastrocnemius or semimembranosus muscles or both. It can be thought of as a chronic knee joint effusion (Fig. 37.14) that herniates between the two heads of the gastrocnemius.

In children the mass is often isolated and asymptomatic and resolves spontaneously. In adults the mass almost

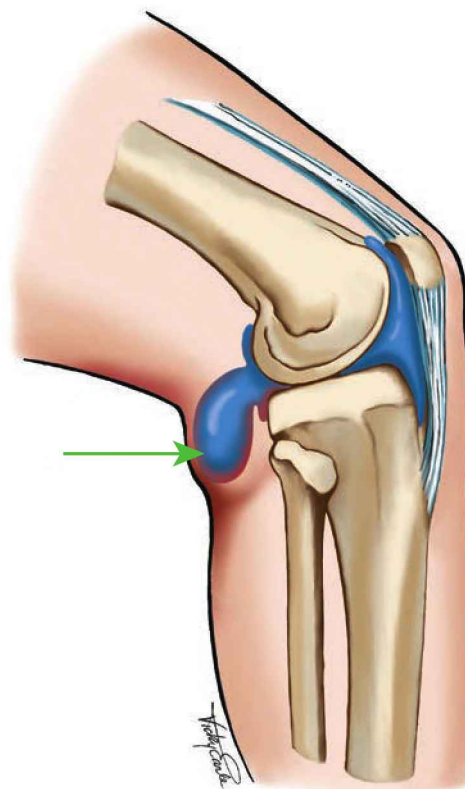


Figure 37.14 Baker's cyst (arrowed)—knee joint effusion herniating posteriorly; usually secondary to degenerative or meniscal pathology

always communicates with the knee joint and is secondary to intra-articular pathology.⁴⁸ Intra-articular pathology includes meniscal tears (most common), ACL deficiencies, cartilage degeneration and arthritis. The size of the mass may fluctuate.

Observation of an athlete's knee in standing often reveals a palpable, swollen, tender mass over the posteromedial joint line. End-of-range knee flexion can be restricted and painful. Deep squats and kneeling may also produce posterior knee pain. Due to its high association with intra-articular pathologies, a full assessment of the knee is warranted.

Radiographs may be of little use in the assessment of Baker's cyst, but may rule out other pathologies (e.g. calcification, loose bodies). Ultrasound can visualise the cyst but cannot assess intra-articular structures, which may be the underlying cause.⁴⁸ MRI will confirm the presence of the cyst and may identify the underlying intra-articular cause—it is considered the gold standard.^{48, 49} Finally, aspiration can differentiate between inflammatory, infectious and mechanical aetiologies. Occasionally the cyst may rupture, leading to lower leg swelling that simulates venous thrombosis. A ruptured cyst usually displays a 'crescent sign'—an ecchymotic (bruised) area around the malleoli.

Initial treatment should involve addressing the underlying cause, (e.g. meniscal tear). Aspiration, together with steroid injection may be useful in the short term.^{48, 50} Surgery may be indicated in symptomatic athletes presenting with a large, symptomatic mass. However, unless the underlying pathology is addressed, it is likely the mass will reform.

Biceps femoris tendinopathy

Biceps femoris tendinopathy has been described earlier under lateral knee pain.

Popliteus tendinopathy

The popliteus muscle arises from the posteromedial border of the proximal tibial metaphysis and travels proximally beneath the lateral head of the gastrocnemius to insert onto the lateral femoral condyle. It also has attachments to the fibula and the posterior horn of the lateral meniscus. A recent cadaveric study has confirmed the role of the popliteus tendon in limiting external and internal rotation, knee varus stresses and anterior translation of the femur (acts with the quadriceps and PCL).⁵¹ Additionally, the popliteus has a role in unlocking the knee during initial flexion from an extended position and aids retraction of the posterior horn of the lateral meniscus to minimise compressive forces during knee flexion.

Clinical features

With its role in knee stability, the popliteal tendon could be thought of as the fifth ligament of the knee.⁵¹ Therefore,

assessment of tibial rotation during a routine knee stability assessment should be considered. Pain associated with the popliteus region may arise from the popliteus muscle, its tendon or the popliteus-arcuate ligament complex. Due to their close proximity, differentiation is difficult.

PRACTICE PEARL

Isolated popliteal muscle injuries are rare. Popliteal muscle injuries associated with lateral meniscus and PCL injuries are more common.^{51, 52}

Posterior lateral instability is often associated with some degree of popliteal pathology.

Twisting activities can result in popliteal injuries, especially with associated posterior knee pain. Impingement of the popliteus tendon can occur with knee rotational instabilities (e.g. following posterior capsule-arcuate ligament strain) and overuse injuries are often associated with repetitive or prolonged acceleration/deceleration activities (e.g. downhill running).

Athletes typically present with posterior knee pain and may report some instability. The main clinical finding is tenderness on palpation along the proximal aspect of the tendon (Fig. 37.13d). With the athlete prone, palpation should begin near the posterolateral corner medial to the biceps tendon and then progress along the medial joint line. Resisted knee flexion in external tibial rotation may reproduce the athlete's pain (Fig. 37.13b).

Garrick and Webb⁵³ describe a test for the popliteus with the patient supine, hips and knees flexed to 90° and the leg internally rotated. The patient is asked to 'hold it there' while the examiner applies an external rotation force (Fig. 37.13c). It is important that the clinician assess both active and passive tibial rotation. Excessive rotation may be a result of repeated strain to the area. Knee flexion and extension range of motion at the end of range may be limited and also reproduce pain. Lower limb biomechanics should be assessed in terms of varus, valgus and rotational forces around the knee.

Soft tissue techniques and mobilisation may improve any restriction of tibial rotation or knee flexion/extension. Rehabilitation should focus on strengthening of the tibial rotators (Fig. 37.15) and hamstring muscles. Any weakness or fatigue in the quadriceps can add excessive strain on the popliteus and should be addressed, for example with a progressive strengthening program that includes tibial internal rotation.⁵⁴ Patients who fail to respond to initial treatment may benefit from corticosteroid injection posteriorly into the point of maximal tenderness or into the popliteus itself, guided by ultrasound.⁵⁵



Figure 37.15 Strengthening the tibial rotators. This may be performed against manual resistance (illustrated), with pulleys, rubber tubing resistance or with an isokinetic machine

Other causes of posterior knee pain

Gastrocnemius tendinopathy

The origin of the medial gastrocnemius at the posterior femoral condyle occasionally is susceptible to painful overuse injuries, especially in runners. This may result from excessive hill running or a rapid increase in mileage.

Examination may reveal local tenderness and associated trigger points in the medial gastrocnemius muscle belly.

Pain may be reproduced on resisted knee flexion, calf raises with the knee in extension, jumping and hopping. Occasionally, stretching the gastrocnemius muscle can reproduce the athlete's pain. Possible biomechanical factors should be assessed, including muscle imbalances around the hip, knee and ankle. Knee stability, footwear and foot function should also be assessed.

Initial treatment may consist of activity modification, ice and local and generalised soft tissue therapy. A graduated/strengthening program is key to good long term results (Chapter 39).

Deep venous thrombosis

Deep venous thrombosis usually presents as calf pain (Chapter 39), but may occasionally present as posterior knee pain. It usually occurs after surgery or following a period of immobilisation. Due to the relative high incidence of asymptomatic deep venous thrombosis, it is important for the clinician to be liberal in the use of Doppler ultrasound or other vascular testing to rule it out, even with minimal suspicion. Distal swelling and some posterior knee pain or calf pain in an athlete who is on oral contraceptives, obese or post surgical, or who has a history of clotting problems, should be tested.

Claudication

Claudication can occasionally present as posterior knee pain. It can occur in young adults, not only in the older person. Popliteal artery entrapment syndrome usually presents as calf pain (Chapter 39).

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e